

Primary Crime Prevention Apps: A Typology and Scoping Review

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Abstract

In the last decade, an array of smartphone apps have been designed to prevent crime, violence, and abuse. The evidence base of these apps has, however, yet to be analyzed systematically. To rectify this, the aims of this review were (1) to establish the extent, range, and nature of research into smartphone apps with a primary crime prevention function; (2) to locate gaps in the primary crime prevention app literature; and (3) to develop a typology of primary crime prevention apps. Employing a scoping review methodology and following Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, studies were identified via Web of Science, EBSCOhost, and Google Scholar. We included English-language research published between 2008 and 2020 that examined smartphone applications designed explicitly for primary crime prevention. Sixty-one publications met our criteria for review, out of an initial sample of 151 identified. Our review identified six types of crime prevention app examined in these publications: self-surveillance apps, decision aid apps, child-tracking apps, educational apps, crime-mapping/alert apps, and crime reporting apps. The findings of our review indicate that most of these forms of primary crime prevention apps have yet to be rigorously evaluated and many are not evidence-based in their design. Consequently, our review indicates that recent enthusiasm over primary crime prevention apps is not supported by an adequate evidence base.

Keywords

apps, crime prevention, digital criminology, smartphones, violence prevention

The smartphone is, today, among the most ubiquitous of digital media—a “radical technology” (A. Greenfield, 2017) that alters the fabric of everyday life. Smartphones are capillaries of hyperconnectivity, enabling users to connect with others through a bevy of mobile- and app-based media. Apps are now central to the fabric of smartphones, so much so that smartphones may readily be dubbed “app-centric media,” to use Daubs and Manzerolle’s (2016, p. 52) term. Apps have not just changed mobile media though. As Kosner (2012) argues, the fabric of the internet is being transformed by appification—the replacing or substitution of websites with apps. Appification, Kosner (2012, np) notes, represents “a fundamental shift in how we metabolize information and entertainment.” Exemplifying this process is the appification of crime prevention: the creation of apps designed to prevent crime through a variety of measures. Crime reporting, personal safety, rehabilitation, and reintegration have all been appified. Consequently, apps with a broad crime prevention focus have received attention from policy makers as a means of preventing various forms of violence and abuse, including intimate partner violence, sexual assault, and child abuse (Colbeck, 2019; Mississippi Department of Child Protection Services, 2020; Sebelius, 2011).

In this article, we conduct a scoping review of research into primary crime prevention apps: apps targeting criminogenic conditions in the physical and social environment and marketed to potential victims of crime. Our review has three main aims:

- To establish the extent, range, and nature of research into primary crime prevention apps;
- To locate gaps in the existing literature into primary crime prevention apps; and
- To develop a typology of primary crime prevention apps.

One of the benefits of examining smartphone-related applications of crime prevention specifically, as opposed to digitally facilitated crime prevention strategies more broadly, is that it enables us to examine the extent to which current crime prevention smartphone apps employ the specific affordances—that is, the possibilities for action—that smartphones afford users (Hutchby, 2001). Smartphone apps are just one instantiation of crime prevention facilitated by digital technologies, and examining smartphone apps enables us to establish whether crime prevention apps are making use of the full possibilities offered by this ubiquitous technology. The smartphone’s

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specific *affordances* have the potential to add value to crime prevention initiatives and practices. However, whether current crime prevention apps are adding tangible value has not yet been subject to systematic empirical inquiry.

Theorizing the Appification of Crime Prevention

At first glance, the appification of crime prevention might appear to be simply another example of the general process whereby a wide range of services, products, and activities have been translated into mobile digital delivery systems. Unlike some human service sectors, justice services have been relatively slow to move into the digital space, and where this has occurred, it has been concentrated in a few areas like law enforcement (Strom, 2016), surveillance, and some correctional treatment and education services (Wardrop & Sheahan, 2019). However, not all digital developments represent genuinely new methods or services or fully make use of the capabilities of mobile digital systems.

Drawing upon Rogers's (2013) distinction between digitized methods and natively digital methods, we can distinguish between digitized crime prevention techniques, digitally enhanced crime prevention techniques, and natively digital crime prevention techniques. Digitized crime prevention methods are predating digital technologies that have simply migrated or been transcoded into digital domains. Natively digital crime prevention techniques, on the other hand, are born out of the specific affordances of the digital—remove the computational technology, and the crime prevention technique cannot exist. Natively digital crime prevention techniques, in other words, always rely on what Floridi (2013, p. 113) terms third-order technologies, which sit between one technology and another. In addition to these two categories, we might speak of crime and violence prevention techniques that are digitally enhanced, that is, rendered significantly more effective through their digitization (Muggah & Macedo Diniz, 2013).

The appification of crime prevention has capitalized on the capabilities of mobile digital smartphone technology, and we can be even more specific in designating whether this technology is a *necessary condition* of a particular crime prevention method. Echoing Wall's (2007) distinction between cyber-enabled and cyber-dependent cybercrimes, we can therefore distinguish between smartphone-enabled, smartphone-enhanced, and smartphone-dependent crime prevention techniques. Smartphone-enabled crime prevention refers to crime prevention techniques that can use smartphones, but do not require smartphones, and are not enhanced by being incorporated into one. Smartphone-enabled crime prevention practices can readily be performed without smartphones—the smartphone represents just one channel through which the crime prevention technique could be performed. However, in many cases, smartphones will improve the speed, range, or effectiveness of the crime prevention technique. In such cases, we might speak of smartphone-enhanced crime prevention techniques.

Smartphone-dependent crime prevention can only be performed using smartphones or mobile technologies with a similar functionality (e.g., mobile-connected tablets). They depend on the mobile affordances of smartphones to undertake the crime prevention technique and, as such, are considerably rarer than smartphone-enabled (or enhanced) crime prevention apps and techniques. Here, it is also worth acknowledging that many of these techniques do not require a specific crime prevention app to use. Rather they can readily be employed using the basic communicative features of a smartphone or a social media app such as Facebook.

In considering smartphone-enabled and smartphone-dependent crime prevention, we need to consider the specific affordances smartphones have that desktop computers and other communicative media lack (Hutchby, 2001). For the most part, such affordances stem from the portability or mobility of smartphones and mobile devices. Like earlier mobile phones (see Cooper, 2001), we can consider the mobility of smartphones as a three-way process that involves the mobility of the user, the mobility of the device, and the mobility of the content. One such affordance generated by this three-way process of mobility is “perpetual contact” (Mascheroni & Vincent, 2016)—the ability to communicate with others anytime and anywhere (Katz & Aakhus, 2002). It is this affordance of perpetual contact that underpins the personal safety and crime alert apps discussed below.

Similarly, the geo-tracking affordances of smartphones can allow for new ways of responding to crime by providing real-time locational information and crime avoidance messaging. Crime avoidance messaging is, of course, not limited to smartphones; police and news media regularly warn people if a dangerous event is currently in progress. However, the mobility of the smartphone serves as an enhancer, in that it allows users to experience notifications about a nearby crime “anywhere, anytime,” and the geo-tracking affordance of a smartphone allows this process to be sensitive to an individual's movements.

One of the distinguishing features of the spread of mobile digital technology is the speed with which new platforms and apps are developed, adopted, and, in some cases, discarded. The spread of crime prevention apps exemplifies this process. None of the apps reviewed in this study existed more than 10 years ago, and most have been developed in the last 5 years. In order to respond effectively to these developments, crime prevention practitioners need to make informed assessments and decisions about the acquisition, adaptation, creation, and use of apps. **At present, we lack a commonly agreed language to describe and classify these digital tools, and there are limited opportunities to share knowledge and experiences about their use. The evidence base to inform assessments of quality or likely outcomes is sparse and unsystematic,** and there are no agreed standards as to what constitutes a good crime prevention app. This scoping review is intended to contribute to generating a common language about crime prevention apps and to map the practice and evidence landscape that users need to navigate their way through.

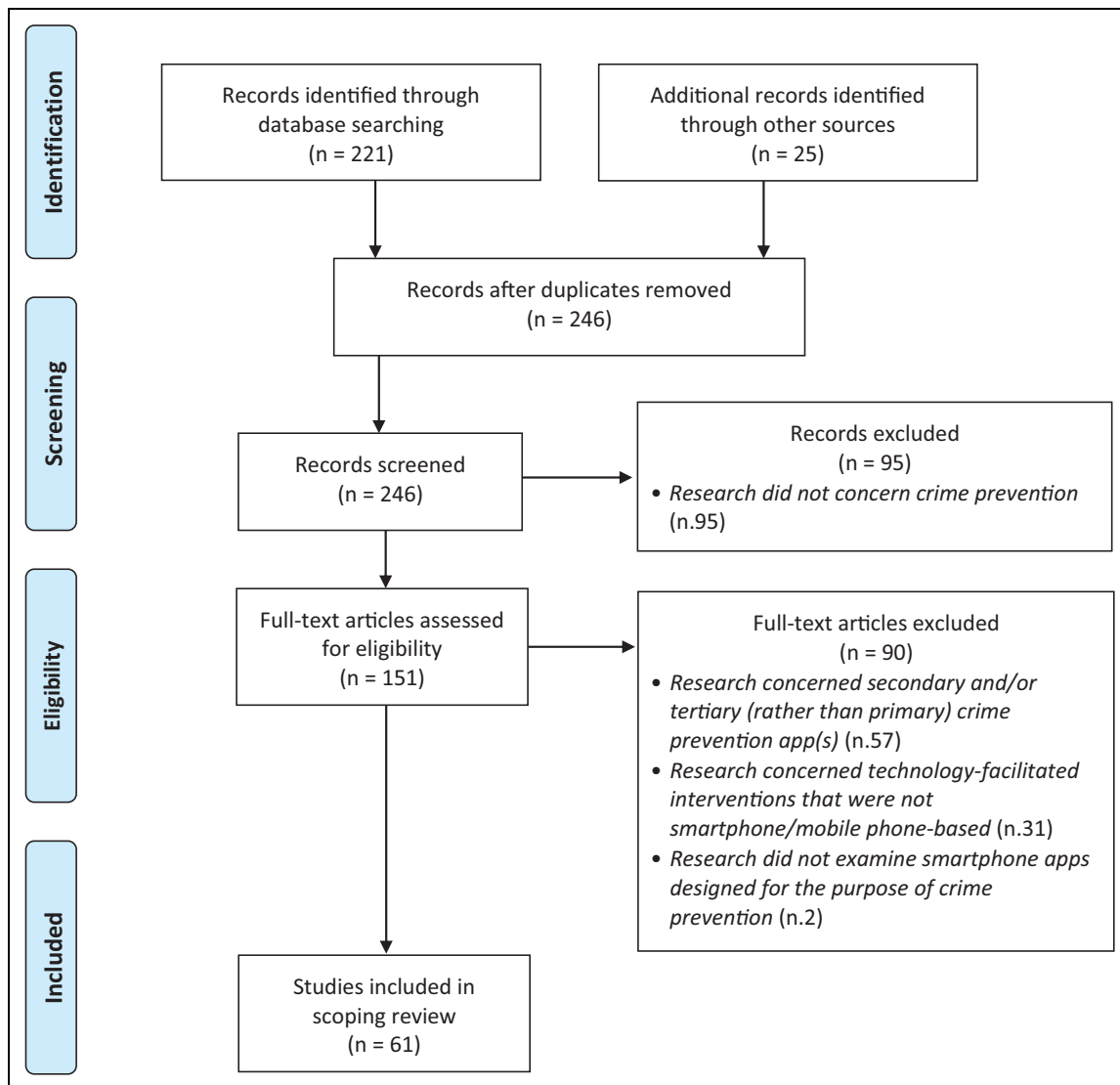


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses diagram summarizing selection of studies for the scoping review. Source. Adapted from Moher et al. (2009).

Method

The articles presented in this scoping review were identified following Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009; see Figure 1). The majority of articles were sourced via **three electronic research databases**: Web of Science, EBSCOhost, and Google Scholar. Articles and conference papers were located using the **following search terms**: “App,” “Apps,” and “Smartphone application” in combination with the following key terms: “access to justice,” “crime,” “crime prevention,” “violence prevention,” “rape prevention,” “justice,” and “personal safety.” **A snowballing approach was then employed to locate additional articles** through scanning the reference lists of each article. As a scoping review (Arksey & O’Malley, 2005), we were concerned less with assessing the quality of included studies than in mapping the key concepts, issues, gaps

in and extent of current research into crime prevention smart-phone apps. For this reason, we included not only peer reviewed journal articles in our review but also unrefereed gray literature such as conference proceedings. Articles between 2008 and 2018 were collected, read, and thematically charted to develop a typology of crime prevention apps and establish the extent, range, and nature of research into such apps.

Scope of This Review

In their influential 1976 conceptual model of crime prevention, Brantingham and Faust (1976, p. 284) noted that the term is “applied confusingly to a wide array of contradictory activities.” Drawing on Brantingham and Faust’s (1976) tripartite model of crime prevention, the focus of this review is on apps that are intended to have a *primary crime prevention*

function. Such *primary prevention apps* seek to modify or abate the social or environmental conditions that give rise to crime or allow potential victims to manage their exposure to risky environments or circumstances.

We have excluded secondary and tertiary crime prevention apps from this review—that is, apps that are intended to modify the behavior of individuals at risk of engaging in crime (secondary crime prevention apps), and “offender-facing” apps intended to treat or manage offending behavior directly, or the behavioral, substance dependency, mental health, or other conditions that accompany offending (tertiary crime prevention apps).

Our reasons for excluding secondary and tertiary crime prevention apps from this review are several. First, all the apps we examined are “victim or consumer facing” and are intended to be used by individuals without any formal connection to justice institutions. Secondary and tertiary crime prevention apps are often forensic versions of existing eHealth or eMental Health apps. Examples include *HealthCall-S*, a self-monitoring and motivational interviewing app for people with substance abuse issues; *eRecovery*, a drug relapse prevention app; the *Digital Anger Thermometer*, designed to monitor and suggest possible intervention for people with anger issues; and *Heart2HAART* that provides automated medication reminders. Both secondary and tertiary crime prevention apps are often intended to operate within or as adjuncts to clinical or institutionally based treatment and support (Morris & Graham, 2020; Ross, 2018). The development processes and operating characteristics of most secondary and tertiary crime prevention apps are thus not comparable with those of primary prevention apps. Moreover, unlike primary prevention apps, nonforensic versions of secondary and tertiary crime prevention apps have already been subject to scoping and systematic reviews (Capon et al., 2016; Kip et al., 2018; Wang et al., 2018; Weaver et al., 2013).

Sample

Our inclusion criteria were as follows: Articles were included if they (1) were published between 2008 and 2020, (2) examined or detailed smartphone applications designed explicitly for primary crime prevention, and (3) were written in English. These inclusion criteria were *deliberatively broad* for two interlinked reasons. In line with the aims of our scoping review, keeping our inclusion criteria broad enabled us to scope out the extent, range, and nature of research into primary crime prevention apps. Further, keeping our inclusion criteria broad enabled us to examine research concerning all phases of the life cycle of these apps, including, importantly, their design. Here, it is worth emphasizing that *crime prevention apps are not only evaluated and examined by researchers; they can also be designed by researchers*. To truly account for the multidisciplinary breadth of scholarship into crime prevention apps, a scoping review must detail not only social scientific studies evaluating the efficacy of these apps; they must also examine how researchers in engineering and computer science studies have approached designing such apps, including the theories

and assumptions that informed their app design. For this reason, we have included in our review software proposals from researchers working in computer science and software design disciplines.

During our review, 151 papers and other academic works were initially sampled. Of these initially sampled papers, 31 were excluded from the sample as they concerned technology-facilitated interventions that were not smartphone/mobile phone based (e.g., interventions that were broadly internet-based but not packaged in a smartphone app form or optimized for smartphone use). Further, two papers initially sampled were removed as they did not examine smartphone apps designed for the purpose of crime prevention. Finally, we excluded 57 papers concerning secondary and/or tertiary crime prevention apps, many of which were mHealth interventions in the form of drug and alcohol treatment and relapse prevention apps. Although not included in this review, we want to emphasize that there is a growing evidence base for mobile mental health apps, detailed in systematic reviews by Rathbone and Prescott (2017) and Wang et al. (2018; see also Seko et al., 2014).

In total, 61 academic works were included in this review: 34 peer reviewed articles, 17 conference papers, seven book chapters, and three university research briefings. Included articles were categorized based on their study design and the discipline of the journal in which they were published. Articles included in our review fell into the following study design categories: experimental studies ($n = 3$), quantitative cross-sectional studies ($n = 12$), qualitative studies ($n = 9$), mixed method ($n = 2$), software proposals ($n = 22$), conceptual analyses ($n = 12$), and scoping reviews ($n = 1$).

Analysis

Articles were independently coded by the three authors and a research assistant using a combination of inductive and deductive thematic analysis (Fereday & Muir-Cochrane, 2006). Any discrepancies in our coding and analysis were discussed and resolved by consensus between the three authors (Syed & Nelson, 2015). Articles were thematically analyzed along the following lines to generate our typology and assess the evidence base of each form of primary crime prevention app we identified:

- *App features*: To generate our typology and examine the extent to which distinct types of primary prevention apps drew upon the specific affordances of smartphones, the features and functionality of the app(s) an article examined were identified and classified using the smartphone enabled, smartphone-enhanced, smartphone-dependent distinction discussed above.
- *Study design*: To assess the evidence base of the different types of primary prevention apps we identified, the study design of articles was cross tabulated with the type of crime prevention app they examined.
- *Crime prevention efficacy*: To assess the efficacy of different types of primary prevention apps we

identified, we examined, first, whether an article studied the efficacy of the app it concerned, and second, whether it found the app to be effective, ineffective, or mixed in its preventative efficacy.

- **Advantages and disadvantages:** Finally, to scope the benefits, disadvantages, and unintended consequences of different types of primary prevention apps, key findings on these issues were cross tabulated with the type of crime prevention app an article examined.

Limitations

Before detailing our results, two key limitations of our review must be considered in interpreting our review's findings. First, our review identified relatively few studies that systematically evaluated apps or tested their preventative efficacy. A larger percentage instead detail design innovations in the crime prevention app industry or examine the unintended consequences and ideological implications of specific types of crime prevention apps. Consequently, our review was able to provide relatively few insights on whether primary crime prevention apps "work" in their stated aim of preventing crime. Although this scoping review was not concerned solely with establishing the evidence base of specific forms of crime prevention apps, this relative lack of studies testing the preventative efficacy of primary crime prevention apps is itself a key finding of our review as it speaks to the need for further research into this issue.

Second, as a result of our search terms and inclusion criteria, our review excluded more general smartphone applications that might be reappropriated to engage in crime prevention. Many general social media apps, including Facebook and WhatsApp, can readily be employed for the purpose of crime prevention (see Pridmore et al., 2019; Wood & Thompson, 2020). For some readers, this might present an issue given that the crime prevention *techniques* employed when people use general social media platforms in this way do not always differ significantly from the crime prevention techniques utilized by apps designed specifically for this purpose. However, we think there is value in examining apps designed specifically for crime prevention given that they occasionally employ crime prevention techniques that are not achievable through general social media applications. Further, there is value in focusing specifically on apps designed specifically to facilitate primary crime prevention given the increasing endorsement of such apps by police forces and universities. Mindful of this limitation of our review though, future research might examine whether and when apps designed specifically for crime prevention have a higher preventative efficacy than social media apps that have been reappropriated for the purpose of crime prevention.

Findings: A Typology of Crime Prevention Apps

We identified six types of apps marketed to potential crime victims and that target criminogenic conditions in the physical

and social environment. These were: self-tracking apps (24 studies), child-tracking apps (three studies), crime-mapping/alert apps (17 studies), decision aid apps (seven studies), crime reporting apps (14 studies), and educational apps (four studies).

Note that our typology is one of ideal types; some crime prevention apps overlap two or more of these categories. As detailed in Table 1, the amount of the research examining each of these forms of apps as crime prevention varies significantly. Child-tracking apps, for example, were the primary focus of only three articles identified by our review. In the following section, we detail the literature examining each of these six types of primary crime prevention apps.

Self-Tracking Personal Safety Apps (n = 24)

The first subset of smartphone apps with a primary prevention intention we identified in our sample were self-tracking apps that aimed to promote the personal safety of users (see Maxwell et al., 2019) by reducing their risk of being the victim of a violent crime. As we detail below, however, the apps researched in this subset of literature varied in the crime prevention techniques they employed and the specific forms of violent offending they sought to counteract.

Many of the personal safety apps discussed in the literature aim to keep users safe through (a) enabling friends or family members to track the user's location, (b) sending an alert to a friend, family member or law enforcement agency in situations where the user feels in danger, and/or (c) automatically sending an alert to family members after a set amount of time if the user does not personally deactivate it. Some of the apps discussed in this literature, such as *Cheeka* (Kanagaraj et al., 2013), were not targeted at a particular demographic. However, the majority of this literature (n = 14) concerned apps designed specifically for women's safety (see Doria et al., 2020; Eisenhut et al., 2020). This echoes Maxwell et al.'s (2019, p. 8) finding that while 19% of the 505 personal safety apps available on the AppStore and Google Play specifically targeted women in their title (e.g., *Be-Safe: The Women Safety App*), or use of terms likely to be more relatable to women, none of the apps they located targeted men specifically. Indeed, Eisenhut et al.'s (2020, p. 6) content analysis of 171 apps designed to address violence against women found apps with an "emergency" (i.e., personal safety) function, were the most common, making up 46.78% of the apps they examined. Further, as Eisenhut et al.'s (2020, p. 6) study indicates, such emergency apps are a mainstay of private sector app development, making up over half the apps designed within the private sector to address violence against women.

There is a notable disciplinary divide in research examining self-tracking apps. One set of literature (n = 8), emanating from the computational sciences and engineering, represent software proposals detailing the features and rationale for apps designed by the authors to promote people's (Sugang et al., 2015), or specifically women's, safety in public space (Chand et al., 2015; Chang et al., 2011; Jiang, 2017; Kanagaraj et al., 2013; Monisha et al., 2016; Yarrabothu & Thota, 2015), or on

Table 1. Study Design, Affordances, Effectiveness, Advantages and Disadvantages Categorized by Type of Crime Prevention App.

App Type	Study Design	Features	Crime Prevention Efficacy	Advantages	Disadvantages
Self-tracking apps ($n = 24$: Beaton 2015; Bivens and Hasinoff 2018; Blayney et al. 2018; Chand et al. 2015; Chang et al. 2011; Doria et al. 2020; Eisenhut et al. 2020; Elcessor 2018; Jiang 2017; Just et al. 2019; Kaims 2017; Kanagaraj et al. 2013; Karusala and Kumar 2017; Liu et al. 2017; Maxwell et al. 2019; Monisha et al. 2016; Natarajan 2016; Potter et al. 2020; Sugang et al. 2015; Viswanath and Basu 2015; Westmarland et al. 2013; Yarrabothe and Thota 2015).	Conceptual analysis: $n = 3$ (Beaton 2015; Elcessor 2018; Kaims 2017). Software proposal: $n = 8$ (Chand et al. 2015; Chang et al. 2011; Jiang 2017; Kanagaraj et al. 2013; Liu et al. 2017; Monisha et al. 2016; Sugang et al. 2015; Yarrabothe and Thota 2015).	Smartphone-enhanced: alarm, machine learning, expert system. Smartphone-dependent: panic button, geo-location tracking, short message service (SMS), camera, iWatch integration, passive triggers for alarm features (including deviating from a set route, changing pace suddenly, or removing headphones from the headphone jack).	No assessment: $n = 21$ (Beaton 2015; Bivens and Hasinoff 2018; Chand et al. 2015; Chang et al. 2011; Doria et al. 2020; Eisenhut et al. 2020; Elcessor 2018; Jiang 2017; Just et al. 2019; Kaims 2017; Kanagaraj et al. 2013; Liu et al. 2017; Maxwell et al. 2019; McCarthy et al. 2016; Monisha et al. 2016; Potter et al. 2020; Sheikh and Fayyaz 2018; Sugang et al. 2015; Viswanath and Basu 2015; Westmarland et al. 2013; Yarrabothe and Thota 2015). Effective: $n = 2$ (Blayney et al. 2018; Natarajan 2016). Ineffective: $n = 1$ (Natarajan 2016). Circle of Six – findings suggest users may not employ the app's interventions in real-world contexts; Karusala and Kumar 2017: panic button apps in New Delhi lacked requisite infrastructural support, did not align with women's values, and participants doubted whether they would have a deterrent effect).	- May increase users' feelings of safety in public space (McCarthy et al. 2016) or on campus specifically (Potter et al. 2020). - May be employed to collect data on factors contributing to fear of crime in (Viswanath and Basu 2015). - Advanced self-tracking apps (such as Companion) enable 'passive' alerts that can prompt alarms or warnings without requiring intentional actions from the user (Elcessor 2018). - Often fail to address the broader social and institutional context of violence [against women], focusing only on individual factors that increase or decrease the risk of experiencing violence (Eisenhut et al. 2020). - May be unfeasible and a safety-risk for individuals who phone share with their spouse – a practice which is not uncommon in low-income and middle-income (Eisenhut et al. 2020). - Public transport journals may lead to 'passive' alert false alarms (Elcessor 2018). - Apps with passive alert features are often not built with disabled individuals in mind, leading to false alarms when users must change their pace, or divert their course to access accessible routes (Elcessor 2018).	- May reinforce victim-blaming attitudes through focusing on victims' rather than offenders' behaviour (Bivens and Hasinoff 2018; Westmarland et al. 2013). - May encourage women's 'safety work' (Bivens and Hasinoff 2018; Westmarland et al. 2013). - App features may be considered redundant in light of existing smartphone affordances (Blayney et al. 2018).
	Quantitative cross-sectional: $n = 8$ (Bivens and Hasinoff 2018; Eisenhut et al. 2020; Just et al. 2019; Maxwell et al. 2019; McCarthy et al. 2016; Natarajan 2016; Potter et al. 2020; Sugang et al. 2015; Viswanath and Basu 2015). Qualitative: $n = 2$ (Karusala and Kumar 2017; Sheikh and Fayyaz 2018). Mixed method: $n = 2$ (Blayney et al. 2018; quantitative surveys + qualitative interviews; Westmarland et al. 2013; quantitative content analysis + qualitative interviews). Scoping review: $n = 1$ (Doria et al. 2020).				
Child tracking apps ($n = 3$: Gabriels 2016; Hasinoff 2017; Simpson 2014).	Conceptual analysis: $n = 2$ (Gabriels 2016; Simpson 2014). Qualitative: $n = 1$ (Hasinoff 2017).	Smartphone-dependent: global positioning system (GPS)-enabled location-based.	No assessment: $n = 3$ (Gabriels 2016; Hasinoff 2017; Simpson 2014).	- May create a situation of over-proximity, in which parents mistake control for care (Gabriels 2016). - May create rather than relieve parents' anxieties over the location of children/family members (Hasinoff 2017). - May infringe on rights of children to privacy (Simpson 2014). - May amplify harmful power dynamics in intimate relationships and be used to facilitate stalking (Hasinoff 2017).	

(continued)

Table 1. (continued)

App Type	Study Design	Features	Crime Prevention Efficacy	Advantages	Disadvantages
Crime mapping/alert apps ($n = 17$): Ali et al. 2015; Ballesteros et al. 2014; Ceccato 2019; Eisenhut et al. 2020; Elcensor 2018; Fabito et al. 2016; Furtado et al. 2010; Furtado et al. 2012; Maghanoy 2017; Moon et al. 2014; Mowlabocus 2018; Rentschler 2018; Revier 2018; Shah et al. 2011; Vaghela and Shih 2018; Weiss 2016; Williams 2016).	Conceptual analysis: $n = 6$ (Elcensor 2018; Mowlabocus 2018; Rentschler 2018; Revier 2018; Weiss 2016; Williams 2016). Quantitative cross-sectional: $n = 4$ (Ceccato 2019; Eisenhut et al. 2020; Moon et al., 2014; Vaghela and Shih 2018). Software proposal: $n = 7$ (Ali et al. 2015; Ballesteros et al. 2014; Fabito et al. 2016; Furtado et al. 2010; Furtado et al. 2012; Maghanoy 2017; Shah et al. 2011).	Smartphone-enhanced: crowd-mapping, area safety scores and/or analytics, cross-confirmation of incidents, user reputation scores.	No assessment: $n = 17$ (Ali et al. 2015; Ballesteros et al. 2014; Ceccato 2019; Eisenhut et al. 2020; Elcensor 2018; Fabito et al. 2016; Furtado et al. 2010; Furtado et al. 2012; Maghanoy 2017; Moon et al., 2014; Mowlabocus 2018; Rentschler 2018; Revier 2018; Shah et al. 2011; Vaghela and Shih 2018; Weiss 2016; Williams 2016).	Apps may make users feel safer in public space (Weiss 2016).	- Often rely on subjective and racially-coded perceptions of what constitutes 'suspicious' behaviour, and consequently can lead to the profiling of racial and socioeconomic groups (Elcensor 2018; Rentschler 2018; Williams 2016). - May foster vigilantism (Williams 2016; Rentschler 2018). - Often fail to address the broader social and institutional context of violence [against women], focusing only on individual factors that increase or decrease the risk of experiencing violence (Eisenhut et al. 2020). - May be unfeasible for individuals who phone share with their spouse – a practice which is not uncommon in low-income and middle-income countries (Eisenhut et al. 2020). - Once users document their experiences on an app, they may lose control of the event's narrative, and may have it harmfully reinterpreted by other users (Rentschler 2018). - Crime-mapping apps drawing upon official databases may draw upon historic, unreliable, and often-out-of-date information (Mowlabocus 2018). - Apps (and especially those including a safe route feature) may give users a false sense of safety (Weiss 2016).
		Smartphone-dependent: GPS navigation device, GPS-based tracking to identify location of victims in need of emergency services assistance, personal navigation assistant, nearby incident warning system.			
Decision aid apps ($n = 7$): Alhusen et al. 2015; Bloom et al. 2016; Doria et al. 2020; Glass et al. 2015; Lindsay et al. 2013; Sabri et al. 2019; Tarzia et al. 2017).	Experimental: $n = 2$ (Glass et al. 2015 [trial protocol]; Sabri et al. 2019 [trial protocol]). Qualitative: $n = 4$ (Alhusen et al. 2015; Bloom et al. 2016; Lindsay et al. 2013; Tarzia et al. 2017). Scoping review: $n = 1$ (Doria et al. 2020).	Smartphone-enhanced: links to resources, personalized messages, risk assessment questionnaire, personalised safety plan, interactive visual aid.	No assessment: $n = 7$ (Alhusen et al. 2015; Bloom et al. 2016; Doria et al. 2020; Glass et al. 2015; Lindsay et al. 2013; Sabri et al. 2019; Tarzia et al. 2017).	- Information is constantly available to smartphone users owing to the mobility of the device (Tarzia et al. 2017). - Can provide information tailored to the needs of an individual (Lindsay et al. 2013). - Perceived to be an appropriate technology-facilitated response to intimate-partner violence by victim-survivors (Bloom et al. 2016; Lindsay et al. 2013). - Enables victim-survivors to access help without fear of judgment or having their children removed (Tarzia et al. 2017).	May be unfeasible and a safety-risk for individuals who phone share with their spouse – a practice which is not uncommon in low-income and middle-income countries (Bloom et al. 2016; Eisenhut et al. 2020; Lindsay et al. 2013; Tarzia et al. 2017).

(continued)

Table 1. (continued)

App Type	Study Design	Features	Crime Prevention Efficacy	Advantages	Disadvantages
Crime reporting apps ($n = 14$: Agangiba and Agangiba 2013; Chand et al. 2014; Charitou et al. 2019; Eisenhut et al. 2020; Fabito et al. 2017; Ferreira et al. 2020; Fidow et al. 2019; Kersten et al. 2019; Mendel and Sharapov 2020; Mwiya et al. 2015; Oduor et al. 2014; Rentschler 2018; Roshan et al. 2017)	Software proposal: $n = 8$ (Agangiba and Agangiba 2013; Chand et al. 2014; Fabito et al. 2017; Fidow et al. 2019; Mwiya et al. 2015; Oduor et al. 2014; Paterson et al. 2018; Roshan et al. 2017). Conceptual analysis: $n = 3$ (Charitou et al. 2019; Kersten et al. 2019; Rentschler 2018). Qualitative: $n = 1$ (Ferreira et al. 2020). Quantitative cross-sectional: $n = 2$ (Eisenhut et al. 2020; Mendel and Sharapov 2020).	Smartphone-enhanced: links to resources and/or support services, anonymous encrypted tip submission, real-time chat, crime trends analysis, cross-validation of incidents; crime mapping; ability to store reports of repeated, related incidents. Smartphone-dependent: guided recall, global positioning system (GPS).	No assessment: $n = 14$ (Agangiba and Agangiba 2013; Chand et al. 2014; Charitou et al. 2019; Eisenhut et al. 2020; Fabito et al. 2017; Ferreira et al. 2020; Fidow et al. 2019; Kersten et al. 2019; Mendel and Sharapov 2020; Mwiya et al. 2015; Oduor et al. 2014; Paterson et al. 2018; Rentschler 2018; Roshan et al. 2017).		App features may be considered redundant in light of existing smartphone affordances (Blayney et al. 2018).
	Experimental: $n = 1$ (Moon et al. 2017). Qualitative: $n = 1$ (Ragavan et al. 2020). Quantitative cross-sectional: $n = 2$ (Eisenhut et al. 2020; Mendel and Sharapov 2020).	Smartphone-enhanced: app-based education program.	No assessment: $n = 3$ (Eisenhut et al. 2020; Mendel and Sharapov 2020; Ragavan et al. 2020) Effective: $n = 1$ (Moon et al. 2017).	Information is constantly available to smartphone users owing to the mobility of the device (Ragavan et al. 2020).	- Smartphone-enhanced nature of the prevention technique may not warrant the creation of a standalone app when a standard website may suffice. - May be unfeasible and a safety-risk for individuals who phone share with their spouse – a practice which is not uncommon in low-income and middle-income countries (Eisenhut et al. 2020; Ragavan et al. 2020).
	Education apps ($n = 4$: Eisenhut et al. 2020; Mendel and Sharapov 2020; Moon et al. 2017; Ragavan et al. 2020).				

university campuses specifically (Liu et al., 2017). Focused on the functionality and usability of their proposed apps, none of these proposals for personal safety apps draw explicitly on crime prevention literature or research into offender behavior (Chand et al., 2015; Liu et al., 2017).

In seeking to promote users' personal safety, many of these apps employ the affordances of global positioning system (GPS) tracking, offering properly smartphone dependent techniques of crime prevention. Liu et al.'s (2017) proposed campus security app, for example, uses Wi-Fi fingerprinting localization combined with Bluetooth beacon-based trilateration to produce a more location-aware app that is able to locate the exact indoor position of a user in distress. Another example is *iServe*, a proposed app which would use an expert system and machine learning to offer an artificial intelligence supported emergency response system (Sucgang et al., 2015). Finally, Viswanath and Basu's (2015) research on *SafetiPin* examines a personal safety app aimed at not only providing safety information for consumers but also directing government bodies to public safety concerns and areas for improvement. Additionally, though not an app *per se*, but rather an adapted technological interface, Vodafone's *TecSOS* phone operates on a similar principle to personal safety apps that aim to offer users digitally-enhanced access to police in cases of emergency. In an evaluation the London Metropolitan Police Service' use of *TecSOS*, Natarajan's (2016) findings indicated that the technology increased the apprehension of offenders and lowered the risk of repeat victimization among users.

Several of the articles detailing self-tracking personal safety apps demonstrate a form of technological solutionism (Morozov, 2013) in framing violence against women as a social issue that can be "solved" through the effective use of technology. Section 3 of Chand et al.'s (2015) conference presentation on their proposed app, *WoSApp* (Women's Safety App), is entitled "Problem and Solution Description," while elsewhere they state: "With the number of criminal acts towards women increasing at such an appalling rate, it is evident that a method is required from the technical community to ameliorate the situation."

The assumptions behind this techno-optimism and solutionism are questioned in the second, critical set of literature into personal safety apps emanating from the social sciences and public health, which questions the preventative efficacy (Maxwell et al., 2019; Karusala & Kumar, 2017), uptake (Blayney et al., 2018), and ideological presuppositions (Beaton, 2015; Bivens & Hasinoff, 2018; Kalms, 2017) of such apps. As Westmarland et al. (2013) note, personal safety apps marketed at women promote what Kelly (2017) has termed "women's safety work": "the range of modifications, adaptations, decisions that women take often habitually in order to maintain a sense of safety in public spaces" (Vera-Gray, 2018, p. 14; see also Stanko, 1997). Similarly, in an analysis of the features and messaging of 215 English-language apps designed to prevent sexual violence, Bivens and Hasinoff (2018, p. 15) demonstrate that the features of the majority of such apps "rely on, and in turn promote, problematic rape myths"—for example, that the

majority of sexual assaults are perpetrated by strangers—and place the responsibility for avoiding assaults on victims. Further, as Maxwell et al. (2019) found in their content analysis of personal safety apps available on the Apple AppStore and Google Play, while marketed as promoting personal safety, few of the apps they analyzed were actually *preventative* in nature. Rather, "the majority of apps intervene when the criminal event is already taking place or help to render assistance after it has already occurred," leading them to conclude that while personal safety apps may reduce fear of crime, they are "not necessarily reducing vulnerability to victimization" (Maxwell et al., 2019, p. 13).

Child-Tracking Apps (n = 3)

Although less well represented in the literature than self-tracking "personal safety" apps, another category that draws on smartphones' geo-tracking affordances are apps designed to promote child safety through "other tracking": using geo-tracking to track the location of other individuals, in this case children and family members (Gabriels, 2016). Hasinoff's (2017) research into *Life360*, a family networking app released in 2008, demonstrates how the GPS tracking functions of smartphones have been deployed in apps designed to track family members' locations in real time. *Life360* markets itself as "the complete awareness package" where the family are "safe, synchronized, and together" as a result of the app's affordances for viewing the real-time location of friends or family members on a private map and receiving real-time alerts when friends or family arrive or leave home, work, and school. However, as Hasinoff (2017) warns, such apps arguably capitalize on and reproduce a false premise that the technology industry promotes to sell information as a product: that digital data sets are neutral and objective. Further, Hasinoff (2017, p. 508) argues that location-based surveillance technologies such as *Life360* "can provide a false comfort that, with enough data, the world can be organized into safe and dangerous categories of people, places, and times" (see Simpson, 2014).

Crime-Mapping/Alert Apps (n = 17)

Another subset of "personal safety" apps that have received some attention from researchers are apps that attempt to promote personal safety through (crowd)mapping crime and providing users with alert services (Furtado et al., 2010) and/or information on "safer" routes and locations in real time. Crowdmapping, as Quaintance (2011, np) explains, refers to "the aggregation of crowd-generated inputs such as text messages and social media feeds with geographic data to provide real-time, interactive information on events." Often associated with the mapping of disaster zones ("disaster mapping"; McCormick, 2010), crowdmapping has increasingly come to have relevance to criminology through its use to crowdmap crime for personal safety (Maghanoy, 2017) and law enforcement for countersurveillance (Wood & Thompson, 2018). Like the self-surveillance personal safety apps discussed above,

some of these apps, such as *StreetSafe* (Ali et al., 2015), are targeted at women to combat street harassment and sexual assault.

One notable example of this form of app is the *WikiCrimes* app developed by Furtado et al. (2010, 2012). *WikiCrimes* is a platform that harnesses the power of collective intelligence and anonymous mass collaboration with an aim to facilitate citizen crime prevention, provide more transparency and publicity to information on crime, and reduce the underreporting of crime (Furtado et al., 2010, p. 5). Given their use of crowdmapping, we might, therefore, term this technique “crowd-crime-mapping.” A synthesis of crowd-mapping and crime-mapping, crowd-crime-mapping is one of numerous contemporary crime prevention and investigation techniques that draw upon the natively digital affordances of crowdsourcing (see, e.g., Monahan & Mokos, 2013; Schafer, 2013; Trottier, 2014). Further, in providing situated and often real-time information on risks to users, such apps capitalize on the mobile affordances of smartphone technology.

While many crime-mapping apps provide locational information on all crime types, some apps, such as the *Sex Offender Tracker* app, map only specific types of crime. Here, we can include anti-street harassment apps such as *Hollaback!* (see Rentschler, 2018; Weiss, 2016) as well as apps providing locational information on individuals listed on sex offender registries. Mowlabocus’s (2018) analysis of the *Sex Offender Tracker* app provides a detailed example of the latter, which, unlike many other crime-mapping apps, provides information not on crime events, but on the individuals who have perpetrated them. Created in 2010 and taken off-line by its creators in 2016, *Sex Offender Tracker* provided users with maps “pinning” the home addresses of individuals listed on a sex offender registries as well as information on these individuals including their mugshot, height, weight, and distinguishing body markings. In addition to potentially hampering the reintegration of these individuals, apps such as the *Sex Offender Tracker* draw upon the unreliable and often-out-of-date information of official registries. This means that while the app gives the sense of providing immediate, live data, “what the user views on Tracker is the cartographic representation of historical data” (Mowlabocus, 2018). Unlike many of the other apps that fall in this category, then, the *Sex Offender Tracker* does not employ the technique of crowdmapping; however, the preventative mechanism it proposes is very much in line with the other apps in this category.

Like the other forms of personal safety-focused apps detailed above, much of the literature devoted to such consumer-facing crowd-crime-mapping apps comes from developers within the computer sciences and represents not evaluation studies but gray literature detailing the features of a proposed or recently designed app (Ballesteros et al., 2014; Fabito et al., 2016, 2017; Furtado et al., 2010, 2012; Maghanoy, 2017; Shah et al., 2011). There were, however, two exceptions to this. The first is Ceccato’s (2019) exploratory study of *SafeLand*, an app developed and employed in Sweden to support crime prevention initiatives in the country. Analyzing 9 month

of app trace data ($n = 5,210$) and surveying users ($n = 72$) about their experiences employing the app, Ceccato found that the majority of users surveyed (87%) indicated that they would be prepared to help if an emergency call was issued by other users of the app. Further, Ceccato found that, while the app was marketed as a tool for crime prevention, it was used more commonly for the purposes of crime reporting; more than 60% of the user posts examined reported a crime, while only 36% of posts concerned crime preventative actions such as reporting suspicious behavior.

The second article drawing on original research of a consumer-facing crime-mapping app was Vaghela and Shih’s (2018) research the *Walksafe* campus safety app, which evaluated the app’s warning system efficacy against the Pennsylvania State University’s emergency notification system—PSUAlert. While participants in Vaghela and Shih’s (2018) study rated the *Walksafe* app as favorable in terms of perceived security, willingness to use, and willingness to share with others, it is worth highlighting that their study evaluated users’ perceptions of the app rather than its actual preventative efficacy. In short, whether crime-mapping and alert apps represent effective crime prevention as opposed to merely affective crime prevention that generates feelings of increased safety among users (see Freiberg, 2001) has yet to be rigorously evaluated in any empirical study.

Even if crime-mapping represented an effective form of crime prevention, myriad concerns remain, including the potential for crime-mapping apps to profile racial and socioeconomic groups (Revier, 2020; Williams, 2016). As Ellcessor (2018) puts it, with such apps, “the racial and class biases of users can become those of the technology, and those of the state, all through a naturalization of ‘data’ as information rather than anecdote or affect.” This potential for app-facilitated profiling is well illustrated in the cases of the now-defunct apps such as *SketchFactor* and *GhettoTracker*. On *Sketchfactor*, users could rate the “sketchiness” of an area and report acts of “sketchy” behavior, with the intention of empowering users to avoid harassment (Dewey, 2014). Similarly, *GhettoTracker*—renamed *Good Part of Town* after criticism over its racist and classist name—allowed users to rate the safety of a particular area, with the intention of helping them avoid unsafe areas (R. Greenfield, 2013). Both apps were quickly labeled racist and classist, with the online backlash leading both to be quickly taken down by the creators—in the case of *GhettoTracker/ Good Part of Town*, within 2 days of going live. As Holmes (2013, np) notes, *GhettoTracker* promotes “idea that poor or crime-heavy areas are places to be categorically avoided or shamed.”

Further, as Rentschler (2018) notes in her analysis of the *Hollaback!* anti-street harassment app, when users detail and geo-locate their experiences of street harassment on the app, their posts can be shared, commented on, and otherwise used in ways the original user may not consent to. In other words, once users document their experiences of street harassment on the app, they lose control of the narrative of the event, and may have it harmfully reinterpreted by other users (Rentschler,

2018)—issues associated not just with *Hollaback!* specifically but with many forms of “informal justice” undertaken through social media (see Wood et al., 2019).

Decision Aid Apps (n = 7)

A further subset of research on personal safety apps is focused on preventing intimate partner violence (IPV) through providing personalized information and decision aids: tools designed to assist individuals in making decisions based on high-quality evidence, especially in relation to health care decisions (Doria et al., 2020; Tarzia et al., 2017). Unlike the personal safety apps discussed above, research into this subset often draws on user group research, to ascertain the suitability (Bloom et al., 2016), and/or usability of proposed apps designed to assist women in abusive relationships. Alhusen et al. (2015) and Bloom et al.’s (2016) formative research that led to the creation of the *MyPlan Safety Decision App*, for example, examines the key considerations in designing an app to assist college-aged women (18–24) experiencing domestic violence (DV) as well as supporting friends and family members in learning more about DV and how to best support their friends or family members while keeping themselves safe. Drawing on surveys, focus groups, and interviews with individuals within the proposed app’s target population as well as survivors of IPV, Bloom et al. (2016) and Lindsay et al. (2013) found that among their participants, an app was widely accepted as appropriate and acceptable for addressing IPV. Further, participants in Lindsay et al.’s (2013) study noted that one of the benefits of an app or website-based platform was the privacy it afforded individuals, compared to accessing resources face-to-face from a friend or another individual. Participants flagged the importance of the app being readily disguisable—a finding consistent with numerous other studies of apps designed to address IPV (Bloom et al., 2016; Lindsay et al., 2013; Tarzia et al., 2017).

In fact, given that participants in studies have flagged the importance of any such intervention app being disguisable, this raises the question of whether an app—which remains visible in some form within an individual’s list of applications—as opposed to a website, which individuals can readily remove from their internet history, represents the most suitable solution. This is particularly salient given recent research on technology-facilitated coercive control that has shown that domestic abusers may monitor their partner’s technology use (see Dragiewicz et al., 2018). As with the other crime prevention apps discussed in this article, it is worth querying what specific value smartphones provide in such preventative interventions. Do such interventions, in other words, make use of the mobile affordances of smartphones so as to offer smartphone-enhanced or smartphone-dependent techniques of IPV prevention?

Unlike the largely unevaluated personal safety in public apps discussed above, several apps designed to respond to IPV have been the subject of quantitative evaluation research, including randomized control trials. *MyPlan*, for example, is currently being evaluated in a randomized control trial

study (Glass et al., 2015) as are *weWomen* and *ourCircle* (Sabri et al., 2019).

Crime Reporting Apps (n = 14)

Crime reporting apps allow users to easily report various forms of crime (Agangiba & Agangiba, 2013; Fabito et al., 2017; Mwiya et al., 2015; Oduor et al., 2014). Unlike crime-mapping apps, which also involve users reporting crimes, crime reporting apps do not tend to provide access to user-submitted data on reported events. The functionality of such apps does not, in other words, allow users to track crime within a particular area and change their behavior accordingly. Examples of crime reporting apps include *Crime Stoppers*, the New Orleans’ *Tip Submit* app, the “crowdsourced child abuse reporting app” *Project Jagriti* (Chand et al., 2014), and *Project Spear* (Roshan et al., 2017), an app for reporting human trafficking.

Although crime reporting apps vary in their features, several employ novel and smartphone-dependent features to improve particular elements of the reporting process, including the quality or utility of the information reported. One notable example of the latter is the recently developed *iWitnessed* app, developed by Paterson et al. (2018) at the University of Sydney, Australia. Drawing on memory research, the *iWitnessed* app employs guided recall to improve witness accounts of crimes (Paterson et al., 2018). Researchers have proposed a number of advantages to such apps. When combined with geospatial search features, such apps may increase situational awareness of crime, as a product of the mobile affordances of smartphones (Agangiba & Agangiba, 2013; Oduor et al., 2014).

Crime reporting apps are an area of rapid development, with a variety of police agencies actively supporting their development and use as part of broader community policing approaches (Kersten et al., 2019). Sophisticated forms of crime reporting apps have the potential to improve efficiency through automated crime reporting (Mwiya et al., 2015), and there have been proposals for “hybrid smart crime reporting apps that combine Twitter sentiment analysis with data collected from crimes reported by members of the public” (Fidow et al., 2019). Finally, unlike other modes of crime reporting, crime reporting apps may employ encryption to facilitate the submitting of anonymous reports (Agangiba & Agangiba, 2013). Regarding this last point, Charitou et al. (2018) highlight that crime reporting apps (and we might add, all the different forms of primary prevention apps surveyed in this article) need to comply with data protection legislation, for example, the European Union General Data Protection Regulation. Drawing on the case study of the *TRILLION* crime-reporting app, Charitou et al. (2018) demonstrate that by employing a security by design approach, crime reporting apps can comply to the EU General Data Protection Regulation’s strict requirements without compromising operational efficacy.

Education Apps (n = 4)

This category describes crime prevention apps with the function of providing users with educational materials, including

about recognizing specific forms of crime. Unlike decision aid–based crime prevention apps, discussed above, educational crime prevention apps do not provide individually tailored messages or plans but rather provide users with uniform, generic information about particular crime issues. Given crime prevention agencies' reliance on fact sheets, advisory guides, and public education campaigns via web-based distribution channels, the scarcity of research examining educational crime prevention apps is noteworthy. Further, the comparative dearth of research into education apps is notable, given Eisenhut (2020, p. 4) found that education apps were the second most prevalent type of app designed to address violence against women, making up 21.05% of the 171 apps they examined. In terms of overlap with other categories, some crime prevention apps incorporate educational material as an extra feature: *Circle of 6*, for example, provides information on sexual violence and resources for victim survivors and friends.

Unlike several of the other app categories charted in this review, educational crime prevention apps do not draw upon the specific mobility-related affordances of smartphones. In other words, their techniques are not smartphone dependent but, at best, smartphone-enhanced. Further, such apps can offer a digitally enhanced crime prevention technique if they harness the interactivity of new media through, for example, presenting information in nonlinear ways (see Laurillard et al., 2000).

We identified only three studies examining apps that fell primarily within the educational crime prevention category (Mendel & Sharapov, 2020; K. J. Moon et al., 2017; Ragavan et al., 2020). In the first of these, K. J. Moon et al. (2017) evaluated *SAP_MobAPP*, an app designed for primary school-aged children that aims to teach children to recognize sexually abusive behavior or situations and to improve their ability to avoid and protect themselves from abuse. While, like other educational crime prevention apps, *SAP_MobAPP* makes use of only smartphone-enabled techniques of crime prevention, the results of K. J. Moon et al.'s (2017) evaluation indicate that the app may have utility in educating children about child sex abuse. Although their results were not statistically significant, K. J. Moon et al. (2017) found that participants' skills in (a) recognizing and (b) avoiding sexual abuse increased more among members of their experimental group, who were trained to use the *SAP_MobAPP*, than members of their control group who received a received internet-based sexual abuse prevention education.

In the second study, Ragavan et al. (2020) examined IPV survivors and providers feedback on *Thrive*, an app designed to improve the well-being of mothers who have experienced IPV. In addition to praising the trauma-informed design of *Thrive* and its usability, the survivors and providers interviewed by Ragavan et al. (2020) emphasized the need for safety precautions, such as password protection and a "quick exit" button, to be built into the design of such apps. In this sense, Ragavan et al.'s (2020) findings align strongly with the qualitative studies of decision aid apps designed for survivors of IPV discussed earlier in this article (Bloom et al., 2016; Lindsay et al., 2013; Tarzia et al., 2017).

Finally, Mendel and Sharapov's (2020) content analysis of 63 "anti-trafficking apps" offers a critical perspective on the potential unintended consequence of poorly conceived and executed educational crime prevention apps. Although not all of the apps they examined fall wholly under the umbrella of educational apps, 87% of the apps they examined did have an "awareness raising" function in seeking to educate users about the signs of people trafficking. As Mendel and Sharapov's (2020, p. 10) study found, not only did 75% of the apps they examined have fewer than 100 downloads, but 17% had featured functions that did not work. Equally troubling, Mendel and Sharapov (2020) found that misleading information about the causes and nature of human trafficking was rife among the apps they examined. Taking an agnotological approach and drawing on Pfaller (2017) and Žižek's (2009) concept of interpassivity, Mendel and Sharapov's (2020) study indicates how, rather than educating users about people trafficking, such apps may in fact produce different forms of ignorance about it. As they argue, in such apps, "ignorance facilitates an interpassivity where it is believed that the apps are acting in an individual or organization's place" (Mendel & Sharapov, 2020, p. 4).

Discussion

Until recently, crime prevention was primarily a mediated process—police, local, or state government agencies developed and promoted programs and tool kits to individuals and communities who were encouraged to conduct risk audits, adopt safe behaviors, lock their doors and windows, or be reassured by newly installed closed-circuit television cameras. However, with the ready availability of mobile crime prevention apps, this dynamic has fundamentally changed. People are now able to choose from a variety of mobile apps that claim to offer a range of protections against various forms of mainly violent crime. Some of these apps are created by the traditional providers of crime prevention solutions, some by specialist app developers, and some by entrepreneurial individuals hoping to find a place in the new app economy. Consumers of this new wave of crime prevention apps need to be able to make informed decisions about what they need and how best to use them, and informed choices based on the capacity of alternatives to meet their needs and their use-preferences.

One of the key findings in this scoping review is that there are very large gaps in the product documentation and evidence base for many of the primary crime prevention apps currently available to consumers. For apps distributed via digital app marketplaces, the problem of product documentation is especially acute. Mislabeling of apps is a common issue (Mani et al., 2015; Weaver et al., 2013), with many health and harm-reduction apps providing either incorrect information or promoting potentially dangerous activities. App stores do carry out technical and content reviews; however, the problem of inaccurate and low-quality content in eHealth apps remains a serious issue (BinDhim & Trevena, 2015).

One strategy that has been widely used in relation to eHealth and eMental Health apps is the development of "curated" app

guides, where treatment or support agencies review and select good quality apps and provide advice to prospective users about how to choose an app that is appropriate to their problem. However, this approach requires a comprehensive and relevant set of app attributes or descriptors to guide curators and consumers. The typology described here is one step toward this kind of consumer advice for crime prevention app users.

The second important gap in the crime prevention app literature is the almost complete absence of systematic evaluations of app outcomes. In particular, the efficacy of “personal safety” apps designed to reduce violence in public space has yet to be examined in any form of rigorous evaluation. No studies to date have examined the efficacy of women’s personal safety apps, crime-mapping apps, child abuse prevention apps, and generic personal safety apps in meeting their primary aim: reducing victimization from crime. In addition, few of the papers by designers proposing such apps engaged in depth with research on environmental crime prevention or victimology. This, in part, speaks to the disciplinary context of this work and the need for greater interdisciplinarity in the design of consumer-facing crime prevention technologies.

Crime prevention app development is substantially driven by software engineering considerations, with only limited evidence of a productive dialogue between computational and criminological knowledge in designing apps. Indeed, only seven articles included in our review were published in criminology journals or books (Ceccato, 2019; Charitou et al., 2018; Fidow et al., 2019; Kersten et al., 2019; Natarajan, 2016; Paterson et al., 2018; Revier, 2020). Beyond the computational sciences and engineering, social scientific studies of primary crime prevention apps have overwhelmingly concerned not their efficacy, but their potential unintended consequences. Such studies demonstrate that, rather than examining crime prevention apps in a vacuum, it is imperative that we broaden our gaze to encompass the social, cultural, and political context in which they have emerged and currently operate. Journalists have been quick to express concern over such apps, and the responsibility they seemingly place on individuals to avoid victimization. Rather than altering the behavior of perpetrators, personal safety apps like *Circle of 6*, aim “to limit or alter the behaviour of potential victims” (Zoe Ridolfi-Starr in Zeilinger, 2016, np). Others have also criticized the app’s narrow and hotline-focused information and approach to sexual assault prevention (Wallace, 2012). Crime prevention apps have been criticized, in Garland (2001) and Beck’s (1992) terms, as simply another self-responsibilization strategy, rendering individuals responsible for managing and/or avoiding risks (see Comack & Peter, 2005; Gray, 2009; Mowlabocus, 2018). To use Manovich’s (2001) term, responsabilization has been *trans-coded* into the language and logic of the smartphone, transforming it in the process. Just as it is important to examine the politics of government-funded crime prevention programs (see O’Malley, 1997; Sutton et al., 2008), it is important to examine the politics of crime prevention apps and the forms of subjectivity they assume and (re)produce (Wood, 2020).

Review of the Main Findings and Their Implications for Practice, Policy, and Research

- To date, no studies have evaluated the efficacy of other tracking child abuse prevention apps, crime-tracking/mapping apps, or crime reporting apps. This indicates that recent enthusiasm over these forms of victim-facing app-based crime prevention interventions is not supported by adequate research or evaluation evidence.
- Numerous studies highlight the need for evaluatory research of primary crime prevention apps to consider the potential unintended consequences of such apps, for example, increasing users’ fear of crime or victim-blaming sentiments.
- Few software proposals for primary crime prevention apps grounded their proposed apps in crime prevention theory or evidence. This indicates that there is need for greater dialogue between computational and criminological knowledge in designing such apps, as well as further engagement with studies evaluating which situational and educational crime prevention measures are effective.
- In the absence of systematic reviews of effectiveness, users of crime prevention apps need to receive advice about how to choose an app that will best meet their needs and a realistic appraisal of the likely benefits and risks (including privacy and data security risks) associated with using crime prevention apps.
- In the longer term, crime prevention agencies should carry out systematic user reviews of apps and provide “curated” apps that provide useful crime prevention assistance without compromising users’ safety or security.

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